

Steady-State Thermal Analysis of a Three-Material Composite Bar

A Finite-Element Study in ANSYS: 2-D and 3-D Models

REE 306 — Heat and Mass Transfer

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Abstract

This report presents a finite-element study of steady-state heat transfer through a composite bar made of three metals — bronze (C51000), aluminium alloy and structural steel — joined side by side. Two models are built and solved in ANSYS Mechanical 2021 R2: a two-dimensional (plane) model and a three-dimensional model of the same bar. In both cases the hot face is held at 1000°C , the opposite face at a fixed lower temperature, and the top surface loses heat by convection to the surroundings at 22°C . The models predict the steady temperature distribution, showing a steep gradient near the hot bronze face and progressive cooling towards the steel end, with additional heat removed from the top by convection. The maximum temperature (1000°C) occurs at the heated face in both models, while the computed average temperature is 192.7°C for the 2-D model and 169°C for the 3-D model. The report also discusses the influence of the different material conductivities and of the convection coefficients on the result.

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1 Introduction

Heat transfer by conduction and convection is central to the design of almost every thermal system, from heat sinks and heat exchangers to building walls and electronic packages. When several materials with different thermal conductivities are joined together, the temperature distribution is no longer a simple straight line: each material sustains a different temperature gradient, and any heat lost to the surroundings by convection further reshapes the field. Closed-form solutions exist only for the simplest one-dimensional cases, so numerical tools such as the finite-element method (FEM) — as implemented in ANSYS — are used for realistic geometries and boundary conditions.

1.1 Objective

The objective of this project is to model and solve, in ANSYS Mechanical, the steady-state temperature distribution in a bar assembled from three different metals, and to compare a two-dimensional idealisation with a full three-dimensional model. Specifically the project aims to:

- build the geometry of a three-material composite bar and assign the correct material properties;
- apply fixed-temperature and convection boundary conditions representative of a heated bar cooling to its surroundings;
- solve the steady-state thermal problem in 2-D and in 3-D; and
- extract, present and interpret the resulting temperature fields, and compare the two models.

2 Problem Description

2.1 Geometry

The model is a rectangular bar formed by three equal blocks placed side by side along the x -direction, as sketched in Figure 1. Each block measures 200 mm $\times$ 300 mm; the assembled bar is therefore 600 mm long and 300 mm tall. In the two-dimensional model the bar is treated as a plane body, whereas in the three-dimensional model it is given a depth of 100 mm. From the heated face inwards the materials are ordered **bronze** $\rightarrow$ **aluminium alloy** $\rightarrow$ **structural steel**.

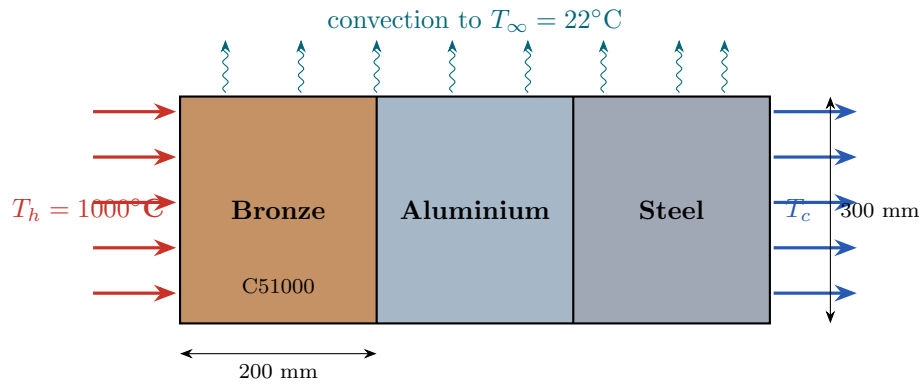


Figure 1. Schematic of the three-material composite bar and its thermal boundary conditions ($T_c = 400^\circ\text{C}$ in the 2-D model, 200°C in the 3-D model).

2.2 Materials

Each block is a different metal. The property that governs steady-state conduction is the thermal conductivity k ; the values taken from the ANSYS material library are listed in Table 1. Aluminium alloy is by far the best conductor, followed by bronze, with structural steel the poorest; this ordering is important for interpreting the results.

Table 1. Thermal properties of the three materials (ANSYS library values).

Material (position)	k ($\text{W mm}^{-1} \text{ }^\circ\text{C}^{-1}$)	k ($\text{W m}^{-1} \text{ K}^{-1}$)	Density (kg mm^{-3})
Bronze C51000 (heated face)	0.0774	77.4	8.72×10^{-6}
Aluminium alloy (middle)	0.144	144–175	2.77×10^{-6}
Structural steel (cold face)	0.0605	60.5	7.85×10^{-6}

3 Finite-Element Model

Both models were built in ANSYS Mechanical 2021 R2 as a *Steady-State Thermal* analysis. The three blocks were bonded at their interfaces (perfect thermal contact), and the assembly was meshed with a program-controlled mesh. The 3-D model naturally contains many more nodes and elements than the 2-D model because it resolves the through-thickness direction as well (Table 2). Figure 2 shows the 2-D mesh.

Table 2. Mesh statistics for the two models.

Model	Nodes	Elements
2-D (plane)	735	210
3-D (solid)	6 339	1 152

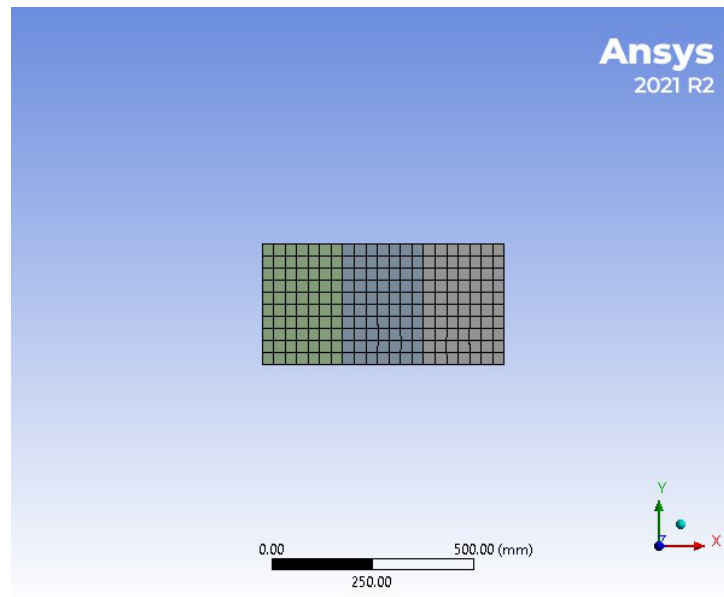


Figure 2. Finite-element mesh of the two-dimensional model.

4 Two-Dimensional Analysis

4.1 Boundary Conditions

The thermal loads applied to the 2-D model are summarised in Table 3 and shown on the geometry in Figure 3. The left (bronze) face is held at 1000°C and the right (steel) face at 400°C . The top edge of each block loses heat by convection to the ambient at 22°C , with a different film coefficient for each block. The initial temperature was 22°C .

Table 3. Boundary conditions of the 2-D model.

Label	Location	Condition
A	Left face (bronze)	Fixed temperature 1000°C
B	Top of bronze block	Convection, $h = 100 \text{ W mm}^{-2}^{\circ}\text{C}^{-1}$, $T_{\infty} = 22^{\circ}\text{C}$
C	Top of aluminium block	Convection, $h = 50 \text{ W mm}^{-2}^{\circ}\text{C}^{-1}$, $T_{\infty} = 22^{\circ}\text{C}$
D	Top of steel block	Convection, $h = 40 \text{ W mm}^{-2}^{\circ}\text{C}^{-1}$, $T_{\infty} = 22^{\circ}\text{C}$
E	Right face (steel)	Fixed temperature 400°C

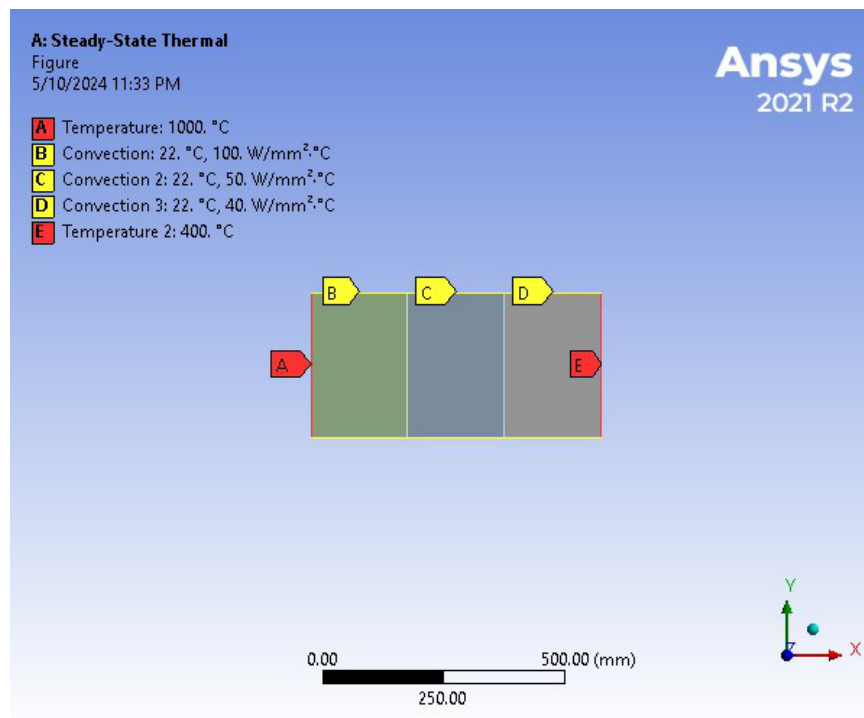


Figure 3. Boundary conditions applied to the 2-D model (A, E fixed temperatures; B, C, D convection).

4.2 Results

The steady-state temperature field is shown in Figure 4 and the extreme values are listed in Table 4. The maximum temperature (1000 °C) occurs, as expected, at the heated bronze face. The temperature drops rapidly through the bar and the coolest region appears near the top, where the strong convection removes heat.

Table 4. Temperature results of the 2-D model.

Quantity	Value (°C)
Maximum temperature	1000
Minimum temperature	−121.13
Average temperature	192.74

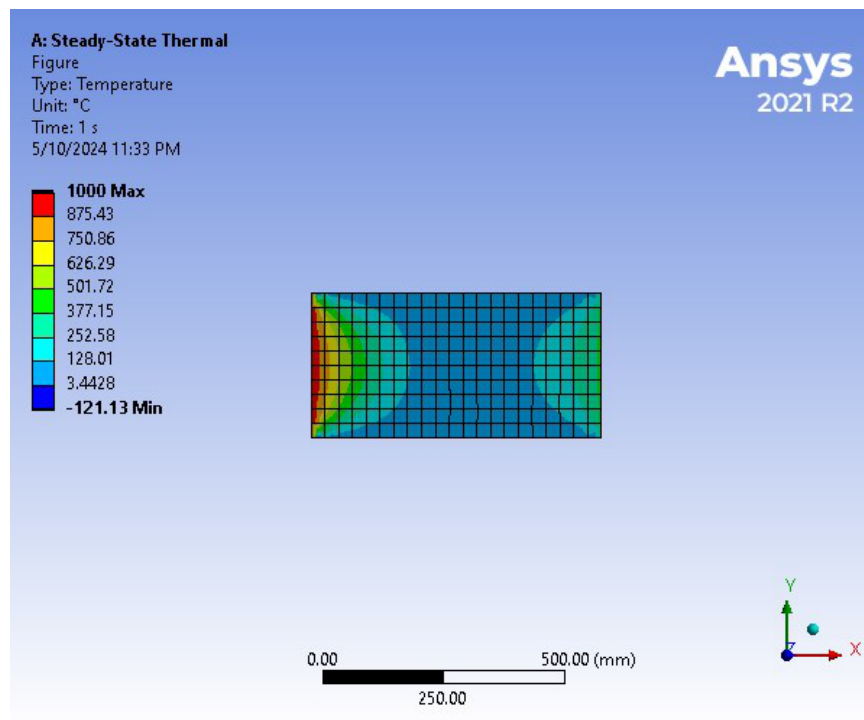


Figure 4. Steady-state temperature distribution in the 2-D model ($^{\circ}\text{C}$).

5 Three-Dimensional Analysis

The three-dimensional model represents the same bar with a 100 mm depth (Figure 5). It uses the same materials and the same type of loading, but with somewhat different numerical values, so that the influence of the boundary values can also be observed.

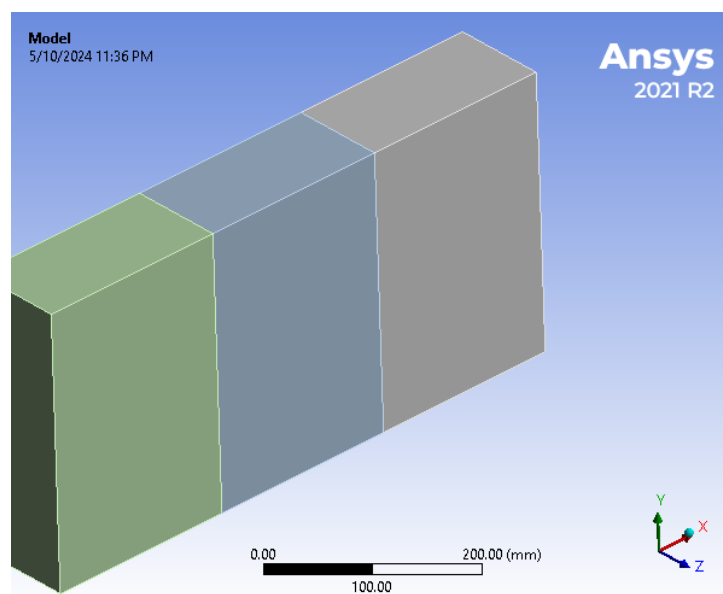


Figure 5. Three-dimensional geometry of the composite bar (depth 100 mm).

5.1 Boundary Conditions

The 3-D loads are listed in Table 5 and shown in Figure 6. The heated face is again held at 1000°C ; the opposite face is fixed at 200°C , and the top faces convect to the 22°C ambient

with higher film coefficients than in the 2-D case.

Table 5. Boundary conditions of the 3-D model.

Label	Location	Condition
A	Left face (bronze)	Fixed temperature 1000 °C
B	Top of bronze block	Convection, $h = 150 \text{ W mm}^{-2} \text{ }^{\circ}\text{C}^{-1}$, $T_{\infty} = 22 \text{ }^{\circ}\text{C}$
C	Top of aluminium block	Convection, $h = 100 \text{ W mm}^{-2} \text{ }^{\circ}\text{C}^{-1}$, $T_{\infty} = 22 \text{ }^{\circ}\text{C}$
D	Top of steel block	Convection, $h = 80 \text{ W mm}^{-2} \text{ }^{\circ}\text{C}^{-1}$, $T_{\infty} = 22 \text{ }^{\circ}\text{C}$
E	Right face (steel)	Fixed temperature 200 °C

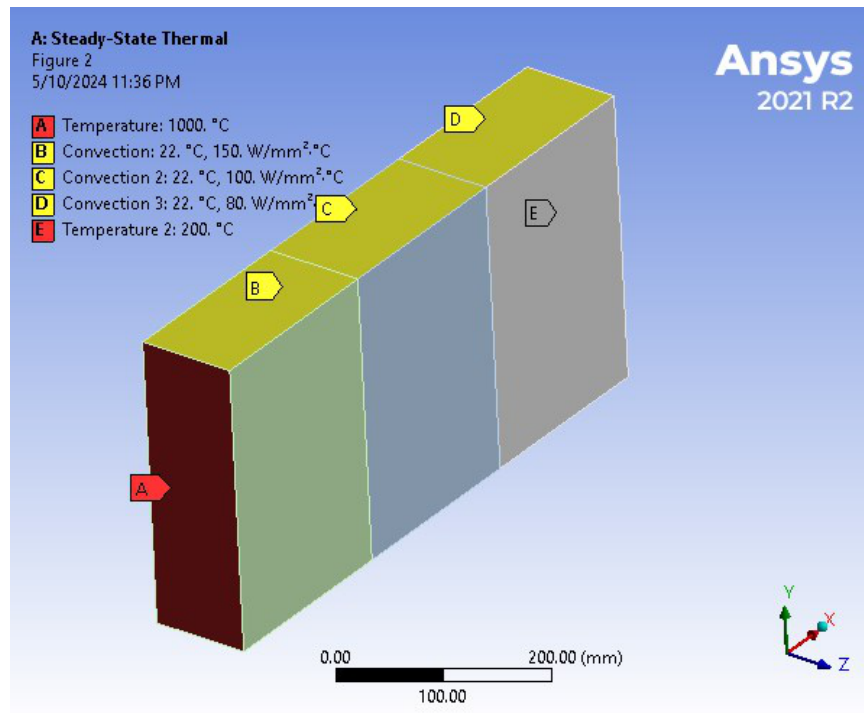


Figure 6. Boundary conditions applied to the 3-D model.

5.2 Results

The 3-D temperature field is shown in Figure 7 from two viewpoints, and the extreme values are given in Table 6. As in the 2-D case the hottest region is the bronze face and the field cools towards the steel end; the contour plots also reveal how the temperature varies through the depth of the bar, information that the 2-D model cannot provide.

Table 6. Temperature results of the 3-D model.

Quantity	Value (°C)
Maximum temperature	1000
Minimum temperature	−121.15
Average temperature	169

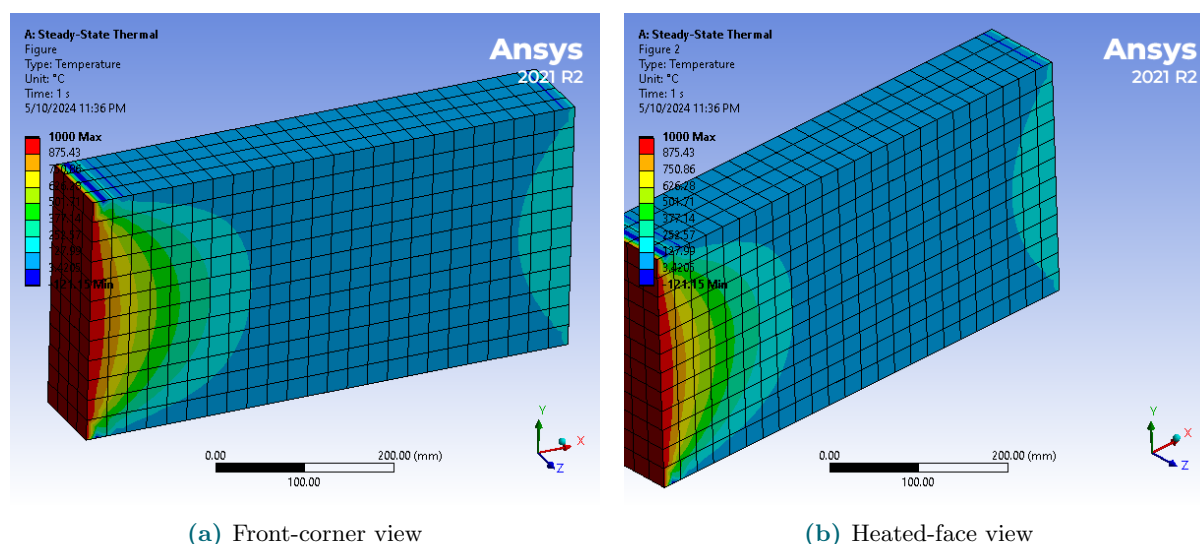


Figure 7. Steady-state temperature distribution in the 3-D model ($^{\circ}\text{C}$).

6 Discussion and Comparison

Table 7 brings the two models together. Several observations can be made.

Table 7. Comparison of the 2-D and 3-D models.

Feature	2-D model	3-D model
Depth	plane (none)	100 mm
Nodes / elements	735 / 210	6 339 / 1 152
Hot-face temperature	1000 $^{\circ}\text{C}$	1000 $^{\circ}\text{C}$
Cold-face temperature	400 $^{\circ}\text{C}$	200 $^{\circ}\text{C}$
Convection h (B/C/D)	100 / 50 / 40	150 / 100 / 80
Maximum temperature	1000 $^{\circ}\text{C}$	1000 $^{\circ}\text{C}$
Average temperature	192.7 $^{\circ}\text{C}$	169 $^{\circ}\text{C}$

- Direction of heat flow.** In both models heat enters at the hot bronze face (1000°C) and flows towards the cooler steel face, while the top surface continuously sheds heat to the surroundings. The steepest temperature gradient is found close to the heated face, which is exactly the behaviour expected from Fourier's law.
- Effect of conductivity.** Because aluminium conducts roughly twice as well as bronze or steel, it sustains the smallest temperature drop across its width; the largest drops occur in the steel block, consistent with its low conductivity.
- 2-D versus 3-D.** The two models give the same maximum temperature but a lower average in 3-D (169°C versus 192.7°C). Part of this difference is genuinely three-dimensional — the 3-D model exposes additional surface area and resolves the through-thickness gradient — and part is due to the different boundary values used (a lower cold-face temperature and larger convection coefficients in the 3-D case). A strict one-to-one comparison would require identical boundary conditions.
- Note on the convection coefficients.** The film coefficients were entered in units of $\text{W mm}^{-2}^{\circ}\text{C}^{-1}$, which are extremely large in SI terms ($1 \text{ W mm}^{-2} \text{ K}^{-1} = 10^6 \text{ W m}^{-2} \text{ K}^{-1}$).

Such intense surface cooling pins the top surface very close to ambient and is the reason the solver reports a small *sub-ambient* minimum temperature ($\approx -121^\circ\text{C}$) near the top corners. For a physically realistic study these coefficients would be reduced to the usual range of tens to a few hundred $\text{W m}^{-2} \text{K}^{-1}$; the qualitative temperature pattern, however, would be unchanged.

7 Conclusion

A three-material composite bar (bronze, aluminium alloy and structural steel) was analysed for steady-state heat transfer in ANSYS Mechanical, using both a two-dimensional and a three-dimensional model. In both cases the temperature field behaves as expected: the maximum of 1000°C occurs at the heated bronze face, the temperature falls towards the steel end, and convection removes heat from the top surface, producing the coolest region there. The average temperature is 192.7°C in the 2-D model and 169°C in the 3-D model, the difference arising from the extra surface and through-thickness resolution of the 3-D model together with its different boundary values. The study demonstrates how material conductivity and surface convection jointly shape the temperature distribution in a multi-material component, and how a 3-D finite-element model provides information — such as the through-thickness gradient — that a 2-D idealisation cannot capture. The main recommendation for future work is to use physically realistic convection coefficients and identical boundary conditions in both models, so that the 2-D and 3-D results can be compared directly.